# NUMBER RACK (REKENREK) ARITHMETIC STUDIO

Dual-Row 20-Bead 5-and-10 Grouping & Subitizing Frame

SERIES 1 • CARD 11 / 12 • DOMAIN: 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

CARD 11 / 12 • RACK-01

MATHEMATICAL GUIDE

## 1. CONCEPT & GEOMETRIC PROOF

### 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

Counting by ones is slow and error-prone. The Rekenrek organizes 20 beads into 5-anchors:
Row 1: 5 Red + 5 White • Row 2: 5 Red + 5 White
Sliding beads in groups turns math into subitizing!

**CALIBRATED 102.0 MM SLIDER DATUM**

- Beads: 10 Red + 10 White = 20 Beads (14.0 mm ⌀)
- Slider Rods: 2x 102.0 mm Octagonal Rods
- Upright Towers: 2x 49.0 mm Snap-Fit Towers (7U)

**5-AND-10 REKENREK: 7 + 6 = 13**




**DOUBLES-PLUS-ONE**

- Red: 5 + 5 = 10 (Instant T
- White: 2 + 1 = 3 Units

**Sum = 10 + 3 = 13**

Zero finger counting!

## 2. KIT PARTS INVENTORY (27 PIECES)

| COMPONENT | QTY | SPEC | COLOR |
|---|---|---|---|
| **Red Beads** | 10x | 14.0 mm Bead | Crimson Red |
| **White Beads** | 10x | 14.0 mm Bead | Classic White |
| **Slider Rods** | 2x | 102.0 mm Rod | Matte Charcoal |
| **Upright Towers** | 2x | 49.0 mm | Matte Charcoal |
| **Storage Tray** | 1x | Nesting Tray | Matte Charcoal |
| **3x1 Storage Bin** | 1x | 49.0 mm | Matte Charcoal |
| **Stage Lid** | 1x | Slider Lid | Matte Charcoal |

**TOTAL: 20 Beads + 4 Frame Parts + Tray + Storage Bin + Lid = 27 Parts**

**DUAL-ROW 5-COLOR STRUCTURE**

- 5 Red + 5 White per row: Instant 5-anchor visual subitizing.
- Both Rows Active: Models doubles, near-doubles, and bridging.
- Zero-Rattle Carry: Beads slide into nested tray slots.
- High Contrast: Crisp color separation across rows.

## 3. PACK-UP & SUPPORT

**✓ 10-SECOND PACK-UP**

Slide 10 red beads left and 10 white beads right on both rods. Place rods and towers into the nesting tray. Missing beads leave a visible gap. Snap lid shut.

**✓ NO LOST PIECE GUARANTEE**

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

**stackcalc.io/parts**
Order replacements anytime

**✓ HANDS-ON TUTORIAL**

Flip over for 5-step concrete lesson & live 3D Video QR!
**Keep stored inside box lid!**

**FUNDAMENTAL THEOREM**

Dual 5-and-10 grouping bridges perceptual subitizing directly into formal arithmetic operations.

**STACKCALC ARCHIVAL INSTRUMENT LABS**
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**OFFICIAL REFILL PORTAL**
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# LESSON GUIDE: DUAL-ROW BRIDGING & DOUBLES PLUS ONE

Step-by-Step Hands-On Tutorial: 7 + 6 = 13 via 5-and-5 Doubles

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LESSON TUTORIAL

## SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/number-rack/

Zero app install • iPad & Chromebook

### CURRICULAR ALIGNMENT

- CCSS 1.OA.C.6 (Addition Fluency)
- CCSS 2.OA.B.2 (Mental Math)
- TEKS 111.3.b.3.C (Compose/Deco...)

Grades K-3 Subitizing & Addition

### GUIDING INQUIRY

When you slide 7 beads on top and 6 on bottom, how many red beads are on the left side altogether?

## 1 STEP 1: SLIDE 7 BEADS ON TOP ROW

Push all beads to the right side of the rack (resting zero).
Slide 7 beads to the left on the top rod in one push:
All 5 red beads + 2 white beads = 7 beads!
Notice: 5-anchor lets you see 7 instantly

## 2 STEP 2: SLIDE 6 BEADS ON BOTTOM ROW

Slide 6 beads to the left on the bottom rod in one push:
All 5 red beads + 1 white bead = 6 beads!
We want to find 7 + 6 without counting one-by-one.

## 3-4 STEPS 3 & 4: GROUP DOUBLES & REGROUP

**3. GROUP RED BEADS: Look at the red beads on the left: 5 + 5 = 10 (Instant Ten)!**
**4. ADD WHITE BEADS: Look at the white beads: 2 + 1 = 3 units.**
Total: 10 + 3 = 13! Therefore: 7 + 6 = 13!
**Calculator Verification: Press Number Rack Subitizing Icon ([≡]) and run [7] [ENTER] [6] [+] → 13.0000.**

## 5 CORE PRINCIPLE: DOUBLES MAKE MENTAL MATH EFFORTLESS

Instead of counting 6 fingers past 7, the Rekenrek decomposes 7 + 6 into (5 + 2) + (5 + 1). The two groups of 5 red beads instantly make 10, leaving 3 white beads. Addition becomes instantaneous visual pattern recognition rather than tedious memorization!

## ★ EXTENSION INVESTIGATION: SOLVE: 8 + 7 USING 5-AND-5

Slide 8 beads on top (5 red + 3 white) and 7 on bottom (5 red + 2 white). Group the 10 red beads first! What is the sum? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.